Mohamed Achich

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PROFESSIONAL SUMMARY

Software engineer experienced in designing and deploying end-to-end full-stack production systems, from user dashboards to backend services and data layers, including hosting and deployment. Skilled in designing modular, role-based architectures for management platforms, applying strong engineering rigor to build scalable and maintainable applications.

Also specialized in broadband device management and IoT ecosystems, with hands-on expertise in TR-069, TR-181, and TR-369 (USP) protocols. Proven experience developing, integrating, and optimizing high-performance embedded systems for reliable, scalable, and secure device communication.

Collaborative team player with a track record of delivering robust solutions in dynamic environments, aligning technical execution with business needs. Passionate about software engineering innovation, continually exploring new technologies to improve system design, automation, and user experience.

PROFESSIONAL EXPERIENCE

Software Engineer — February 2024 – August 2026

PIVA Software, Sfax

ACS/USP Controller Web Platform (SaaS, Full-Stack Web) – React/Node.js

- Designed and developed a multi-protocol TR-369/USP and TR-069/CWMP controller as a **microservices** (**Node.js/Express** backend, **React** frontend, **NestJS**for the authentication service), orchestrated via Docker Compose.
- Developed protocol broker services in **Golang** for **MQTT**, **STOMP** and **WebSocket**, bridging device messages to an event backbone **NATS JetStream** consumed by the backend.
- Implemented a CWMP ACS (TR-069) endpoint handling Inform/RPC processing, sending connection-requests via **UDP**, **HTTP** and **XMPP** to reach CPEs behind **NAT**.
- Designed an authentication and authorization service based on **RBAC** using **Spring Security,** integrating per-organization device credential management via NATS **JetStream KV**, decoupling authentication from the main device management backend.
- Set up automated provisioning of **TLS** certificates for MQTT/STOMP/WebSocket services, and production build pipelines using V8 bytecode compilation for source code protection.
- Worked with **MongoDB** (device/configuration data) and **PostgreSQL** (authentication data) as persistence layers.

Company Website Development & Deployment – React/PHP

- Designed, developed and deployed the official EasyCWMP / PIVA Software website (easycwmp.org), a static **React** frontend paired with a **PHP** backend handling contact form submissions and email notifications.
- Configured and managed end-to-end hosting on IONOS, including domain setup, **DNS**, **SSL/TLS**, as well as production deployment and maintenance.

Web Platform (USP Agent Simulator) – Angular/Java/Spring Boot

- Designed and developed a USP agent simulation web platform using **Angular** and **Java/Spring Boot**, capable of running multiple OB-USP-Agent instances simultaneously for functional and load testing.
- Architected the platform front-end structure (components, services, routing) to enable scalable configuration and control of the simulated agent instances.
- Developed **a real-time monitoring dashboard** within the platform to provide visibility into agent performance, metrics, and system status.
- Containerized the simulator with **Docker**, enabling scalable, reproducible, and isolated test environments.

E-Commerce Shopping Platform – Angular/Java/Spring Boot

- Built a full-stack e-commerce web application using the **Java stack** (MongoDB, Spring Boot, Angular, Java), covering both the customer-facing store and an admin panel.
- Designed and implemented the customer-facing interface, including product browsing, cart, and checkout process, with a **Angular** responsive and dynamic frontend.
- Built the admin panel from scratch to manage products, inventory, and orders, backed by a **REST API** built with **Spring Boot**.
- Designed the **MongoDB** data model and backend architecture to support efficient, scalable management of products and orders.

Business Management ERP – Angular/Java/Spring Boot

- Designed and developed an **ERP web modular application** for small businesses, covering inventory management, invoicing, and employee/HR records as independent yet integrated modules.
- Implemented role-based access control (**RBAC**) with separate portals for administrator, manager, and staff.
- Built an **Angular** dashboard with a **REST API Java/Spring Boot**, using **PostgreSQL** for relational business data (products, stock levels, invoices, employees).
- Containerized the application with Docker for reproducible deployment across environments.

Embedded Android

- Integrated **EasyCWMP**, the TR-069 network management client, into the Android Open Source Project (**AOSP**), validating system functionality, performance, and compatibility.
- Integrated the **USP** (TR-369) agent into the Android environment (**AOSP**), performing functional validation and protocol-level testing.
- Developed a TR-369 USP Controller Android application for **IoT** device management, combining UI/UX design in Figma with a **Kotlin** implementation for scalable, high-performance control.

Broadband Device Management

- Modularized the TR-069 data model by refactoring it into a reusable library and a standalone executable, improving maintainability, scalability, and code reuse.
- Implemented the **Firmware** Activation functionality (Device.DeviceInfo.FirmwareImage.{i}.Activate) at the agent logic and data model level, ensuring compliance with TR-069 specifications.

- Designed and implemented a **caching mechanism** in EasyCWMP, reducing redundant CPE requests and significantly improving performance and efficiency.
- Collaborated with **Luxshare** to design and implement an asynchronous HTTP communication mechanism within the EasyCWMP agent, improving responsiveness and handling of concurrent requests.
- Collaborated with **Tiesse** to map TR-181 data model parameters to a YANG-based data model, ensuring consistency and interoperability between management frameworks.
- Collaborated with **TeamF1** to deploy the ACS/USP controller on their server.

TR-232 Development

- Collaborated with **TCL** to design and implement a **TR-232** Bulk Data Export solution, developing the client-side exporter in C and the collector server in **Golang**.
- Implemented scalable **bulk data reporting mechanisms** to enable efficient telemetry collection from embedded devices, in accordance with TR-232 specifications.
- Contributed to end-to-end system integration, validation, and performance optimization between device and backend components.
- Wrote **detailed technical documentation** covering the system architecture, protocol flow, and implementation design to support deployment and maintenance.

Protocol Integration & Validation

- Collaborated with **Luxshare** to integrate and support **XMPP**-based connection-request mechanisms between **GenieACS** and CPE devices, enabling reliable communication behind NAT.
- Conducted comprehensive protocol testing for TR-369 (USP) using the PMS USP controller, ensuring full functional coverage and reliability, and contributing to validation documentation.
- Conducted comprehensive protocol testing for TR-069 using PMS ACS, ensuring full functional coverage and reliability, and contributing to validation documentation.

Software Engineer Intern — February – May 2024

PIVA Software, Sfax

- Set up the Android Open Source Project (**AOSP**).
- Integrated **EasyCWMP**, the open-source TR-069 network management solution, into the Android environment.
- Adapted **TR-098** and **TR-181** DataModels to the Android environment.
- Developed an Android application for remote device management, improving efficiency.

PROJECTS

Fan League Web Application – React/Node.js — September – March 2026

Freelance

- Designed a modular monolith full-stack architecture (**NestJS** backend, **React** frontend) for a real-time fan engagement platform.
- Designed a dual-ledger points system (competitive XP vs. redeemable loyalty points) using an append-only ledger design to ensure full auditability and prevent leaderboard manipulation.
- Designed **Redis**-based leaderboards using Sorted Sets (season, monthly, daily, all-time, per-team) with sub-second read/write performance and full rebuildability from PostgreSQL as the single source of truth.

- Scheduled **asynchronous processing** (Quartz Scheduler) for score settlement, streak/badge computation, and anti-cheat analysis, to avoid write contention during high-traffic periods.
- Implemented the **authentication service** (Spring Security) with RBAC (Owner/Moderator/Editor) and audit logging, avoiding vendor lock-in and enabling real-time account bans.

Founder & Developer **January 2023 – Present**
Anime Quiz

- Designed and developed an **Android mobile quiz game** targeting anime enthusiasts, offering an engaging and interactive user experience.
- Designed and implemented a modern, responsive interface with intuitive navigation, animations, and optimized user flows to maximize player engagement.
- Applied clean architecture principles and modular development practices to ensure scalability, maintainability, and ease of future feature extension.
- Developed **game logic**, **scoring systems**, progression mechanisms, and question management features to enhance the gaming experience.
- Optimized application performance, resource usage, and responsiveness across a wide range of Android devices.

Embedded Systems & IoT Developer **September 2023 – January 2024**
Green Life Care

- Designed and developed an **IoT-based smart plant monitoring system** and automation based on **STM32 microcontrollers** to improve plant health and resource efficiency.
- Integrated environmental sensors to collect real-time data on light intensity, temperature, and humidity for plant monitoring and analysis purposes.
- Implemented soil monitoring features by integrating an **NPK** sensor with a **STM32**-based system to collect and process soil nutrient data for smart plant management.
- Developed automated irrigation and lighting control mechanisms based on **real-time sensor data** and configurable thresholds.
- Built **a real-time monitoring dashboard** enabling remote visualization of environmental conditions and control of connected devices.
- Integrated **embedded firmware** with **cloud-connected services** to facilitate data collection, device management, and remote access.
- Conducted system testing, calibration, and validation to ensure reliability, accuracy, and stable operation under real-world conditions.

SKILLS

Programming Languages: Golang | JavaScript | TypeScript | C/C++ | Python | PHP | Java
Frameworks & Platforms: Java/Spring Boot | Next.js/React | Angular
Protocols & Tools: Docker | Jira | MQTT | STOMP | WebSocket | Redis | NATS/JetStream | XMPP
Databases: MySQL | PostgresSQL | MongoDB
Operating Systems: Ubuntu | OpenWRT | Fedora | Windows
Device Management Platforms: Genieacs | openacs | PMS USP Controller

EDUCATION

Engineering Degree – Second Cycle **September 2021 – January 2024**
National School of Engineers of Sfax (ENIS)
Engineering Degree – First Cycle **September 2019 – July 2021**
Preparatory Institute for Engineering Studies of Sfax (IPEIS)

CERTIFICATIONS

- **Version Control** – Coursera
- **Scrum Fundamentals Certified (SFC)** – SCRUMStudy
- **Java Spring Boot Backend Development** – Coursera
- **Server-side JavaScript with Node.js –** Coursera
- **Databases and SQL for Data Science –** Coursera
- **Python for Everybody Specialization** – Coursera
- **Linux Shell Scripting Fundamentals –** Coursera